Table 1: Success rate on the factored critic×actor seam: mean over each run's last 3 eval points, ± sd over 3 seeds. **BC** is the `bcflow` control (the Flow actor with `q_coeff`=0, so the critic's value term is removed). Only FB's BC control is available in this source table; because the actor conditions on the critic's conditioning vector (FB's learned z for FB, the goal for the goal-conditioned critics), that row is not the matched control for the GCIQL, biGCIQL and CRL blocks.

		ManiSkill				Metaworld			
		`ms-pick-cube`		`ms-push-cube`		`mw-pick-place`		`mw-push`	
Critic	Actor	expert	play-noise	expert	play-noise	expert	play-noise	expert	play-noise
FB	BC	0.19 ± 0.01	0.57 ± 0.06	0.09 ± 0.05	0.82 ± 0.04	1.00 ± 0.00	0.88 ± 0.03	1.00 ± 0.00	0.93 ± 0.04
	Flow	0.21 ± 0.08	0.69 ± 0.08	0.04 ± 0.04	0.78 ± 0.13	1.00 ± 0.00	0.86 ± 0.03	1.00 ± 0.00	0.96 ± 0.02
	DDPG	0.20 ± 0.03	0.20 ± 0.04	0.10 ± 0.10	0.58 ± 0.08	1.00 ± 0.00	0.91 ± 0.02	1.00 ± 0.00	0.88 ± 0.04
GCIQL	Flow	0.08 ± 0.02	0.14 ± 0.03	0.01 ± 0.01	0.31 ± 0.16	0.38 ± 0.04	0.88 ± 0.06	0.18 ± 0.02	0.94 ± 0.03
	DDPG	0.05 ± 0.00	0.06 ± 0.01	0.00 ± 0.00	0.13 ± 0.05	0.00 ± 0.00	0.58 ± 0.01	0.00 ± 0.00	0.80 ± 0.03
biGCIQL	Flow	0.08 ± 0.03	0.08 ± 0.03	0.00 ± 0.00	0.19 ± 0.06	0.34 ± 0.13	0.85 ± 0.10	0.37 ± 0.24	0.90 ± 0.02
	DDPG	0.06 ± 0.01	0.05 ± 0.00	0.00 ± 0.00	0.16 ± 0.13	0.01 ± 0.01	0.39 ± 0.15	0.10 ± 0.08	0.64 ± 0.05
CRL	Flow	0.05 ± 0.00	0.10 ± 0.04	0.01 ± 0.01	0.43 ± 0.14	0.79 ± 0.07	0.53 ± 0.39	0.64 ± 0.55	0.22 ± 0.10
	DDPG	0.05 ± 0.00	0.06 ± 0.01	0.01 ± 0.01	0.28 ± 0.04	0.40 ± 0.30	0.07 ± 0.04	0.73 ± 0.09	0.10 ± 0.04

Table 2: Pre-registered kill-test arms (phenomenon.tex), launched 2026-08-03. *play-noise-hard* = scripted-oracle mixture with `expert_frac`=0, $\varepsilon_{\max} = 0.5$, $\sigma_{\max} = 0.3$ (reopens headroom, Ph. I only); *play-resample* = OGBench-style play: 200 trajectories × 500 steps, target re-drawn from the env goal space on every success (no dwell) and every 60 steps (equalizes E7: horizon, dwell, goal diversity, kernel range; eval-side goal channel untouched). Cube arms test the Ph. V mechanism. All cells 3 seeds; PENDING = submitted, awaiting results.

Setting	Critic	Actor	`mw-push`	`mw-pick-place`
play-noise-hard (Pred. headroom)	biGCIQL	BC	0.600 ± 0.050	0.533 ± 0.058
	biGCIQL	Flow	$\mathbf{0.867 \pm 0.076}$	0.583 ± 0.058
	GCIQL	BC	0.667 ± 0.029	0.517 ± 0.076
	GCIQL	Flow	0.717 ± 0.076	0.483 ± 0.058
play-resample (Pred. kernel(b))	biGCIQL	BC	0.967 ± 0.029	0.950 ± 0.000
	biGCIQL	Flow	0.983 ± 0.029	0.983 ± 0.029
	GCIQL	BC	0.967 ± 0.029	0.983 ± 0.029
	GCIQL	Flow	0.950 ± 0.000	0.983 ± 0.029
play-resample-hard ($\varepsilon_{\max}$=0.5, $\sigma_{\max}$=0.3)	biGCIQL	BC	0.683 ± 0.126	0.550 ± 0.050
	biGCIQL	Flow	0.700 ± 0.087	$\mathbf{0.667 \pm 0.029}$
	GCIQL	BC	0.600 ± 0.100	0.617 ± 0.115
	GCIQL	Flow	0.650 ± 0.050	0.617 ± 0.076

Cube arm (500k, 3 seeds)	Tests	Success
GCIQL + $\kappa \tanh(\cdot/\kappa)$, κ=75 (`gciql-qb75`)	Ph. V(a) boundedness	0.760 ± 0.037
biGCIQL d=50, no ψ-LayerNorm (`biql-d50-nopsiln`)	Ph. V(b) norm concentration	0.909 ± 0.032
biGCIQL d=15	$d^\dagger$ bracket	0.837 ± 0.027